TXA: timing and regimen

Adults with active/suspected major traumatic bleeding; time from injury.

≤3 h; 1 g IV over ≥10 min → 1 g IV over 8 h Follow institutional policy.

Give early; Chinese R33 states at least eight hours for maintenance, European R23 eight hours. Follow institutional policy.

Beyond three hours, not routine; only with viscoelastic hyperfibrinolysis evidence and senior/local-policy review; avoid known fibrinolytic shutdown. Follow institutional policy.

Follow local product, preparation and infusion policy; avoid rapid IV push.

Chinese shock2023

European bleeding2023

NICE NG39

Follow institutional policy. Medical education and supervised training.

Adult bleeding: BP target

Early adult bleeding before control, without brain/spinal injury.

SBP 80–90 mmHg Follow institutional policy.

Caution with older age/chronic hypertension; separate pediatric/pregnancy protocols.

If targets fail, consider norepinephrine (European R14); Chinese R28 considers it in life-threatening MAP below50 or SBP below70, alongside blood/control rather than replacing them. Follow institutional policy.

Chinese R22 / European R13 support conditional use, not a universal target.

Chinese shock2023

European bleeding2023

Follow institutional policy. Medical education and supervised training.

Shock index

Adult risk recognition, with examination and trends.

SI at least one or pulse pressure below30 mmHg (Chinese R2) Follow institutional policy.

Chinese Table2: SI at least1.0/1.5/2.0 roughly associates with20–30%/30–50%/50–70% loss; beta blockers, age and athletic physiology confound it, so not precise volume. Follow institutional policy.

Blunted HR or apparently normal older-adult vitals do not exclude shock.

European R5 uses SI and/or pulse pressure for shock/transfusion assessment; integrate mechanism, consciousness, perfusion, lactate and response.

Chinese shock2023

European bleeding2023

Follow institutional policy. Medical education and supervised training.

MTP activation assessment

Adults; ACS TQIP example.

ABC ≥2 + clinical state Follow institutional policy.

Penetrating torso injury, positive FAST, HR at least120 and SBP at most90 each score one; ABC at least two triggers assessment; verify local boundaries. Follow institutional policy.

Blood already required in the bay and operative/interventional control are TQIP triggers; consider ongoing bleeding/response without waiting for24-hour10-unit totals. Follow institutional policy.

Initial plasma:platelets:RBC1:1:1–1:1:2 (Chinese R27), then early goal-directed therapy. Domestic RBC units generally derive from200 mL whole blood and differ from overseas units; convert platelet therapeutic amounts with the blood bank. Follow institutional policy.

Initial plasma:platelets:RBC1:1:1–1:1:2 (Chinese R27), then early goal-directed therapy. Domestic RBC units generally derive from200 mL whole blood and differ from overseas units; convert platelet therapeutic amounts with the blood bank. Follow institutional policy.

ACS TQIP massive transfusion guidance

Chinese shock2023

NHC clinical blood use technical standard (2025)

Follow institutional policy. Medical education and supervised training.

Fibrinogen

Major active bleeding or viscoelastic evidence of functional deficit.

Clauss Fib ≤1.5 g/L + major bleeding Follow institutional policy.

European R29 initial fibrinogen concentrate about3–4 g, or cryoprecipitate by measured local content per unit; domestic and overseas units differ, so do not copy15–20 units directly; recheck. Follow institutional blood-bank policy. Follow institutional policy.

Ongoing bleeding/worsening tests: blood-bank/coagulation escalation.

Chinese perioperative consensus uses <1.5 for cryo in severe oozing and <1.5–2.0 for fibrinogen; scope/operators differ.

European bleeding2023

Chinese perioperative

Follow institutional policy. Medical education and supervised training.

Hb target during transfusion

European R16 target for adult trauma requiring RBC transfusion.

Hb 70–90 g/L Follow institutional policy.

Use bleeding/perfusion/trajectory; normal early Hb does not exclude hemorrhage.

TBI, myocardial ischemia and special populations need specific goals; do not await Hb fall.

Chinese 2025 blood-use standard requires individual assessment/response review, not a national 70–90 target.

European bleeding2023

NHC clinical blood use technical standard (2025)

Follow institutional policy. Medical education and supervised training.

Ionized calcium

Adult major bleeding, especially massive transfusion.

iCa at least1.00 mmol/L (Chinese); European normal1.1–1.3 mmol/L Follow institutional policy.

Measure iCa at transfusion start; 10 mL of10% calcium chloride contains about270 mg elemental calcium, versus about90 mg in10% calcium gluconate. Verify concentration/access and recheck. Follow institutional policy.

Ongoing transfusion, falling iCa or rhythm changes: prompt reassessment.

European R31 prefers calcium chloride for hypocalcemia; this compares elemental content rather than interchangeable volume prescriptions. Distinguish the Chinese lower target from European normal range.

European bleeding2023

Chinese perioperative

Follow institutional policy. Medical education and supervised training.

Severe TBI: perfusion target

Severe TBI (GCS ≤8); avoid permissive hypotension.

MAP ≥80 mmHg Follow institutional policy.

For GCS9–15 within three hours, discuss TXA by CRASH-3 scope; TBI bleeding platelet goal above100×10⁹/L (European R30); VHA-guided traumatic-ICH correction (NCS/SCCM2026). Follow institutional policy.

Changing consciousness/pupils or unmet goals: address oxygenation/perfusion and escalate neurological care.

CRASH-3 enrolled GCS at most12 or CT-confirmed intracranial bleeding without major extracranial hemorrhage; subgroup benefit is not universal TBI benefit.

Chinese shock2023

Chinese prehospital2026

European bleeding2023

CRASH-3

NCS/SCCM2026

Follow institutional policy. Medical education and supervised training.

Older adults: occult shock warning

Older-trauma high-risk triage signal, not a stand-alone shock diagnosis.

Age ≥65; SBP <110 mmHg Follow institutional policy.

At least65 years with SBP below110 or HR above SBP raises concern; anticoagulated head injury escalates assessment even with apparently normal signs. Follow institutional policy.

Low-energy injury with symptoms, anticoagulation or poor perfusion: early senior review.

The 110 signal comes from prehospital consensus; hip guidance is a domestic orthopedic pathway entry.

Chinese prehospital2026

NHC geriatric hip fracture care and management guideline (2022)

Follow institutional policy. Medical education and supervised training.

Anticoagulants: reversal pathway

Life-threatening/uncontrolled bleeding or urgent high-risk surgery; authorized team decisions.

Life-threatening bleeding: drug, last dose, INR/renal function/anti-Xa

Warfarin: 4F-PCC25–50 IU/kg by INR plus vitamin K1 5–10 mg slow IV; available 4F-PCC preferred to FFP. Dabigatran: idarucizumab5 g IV. Xa agents: consider4F-PCC25–50 IU/kg; 2026 ICH conditional guidance favors4F-PCC. Follow institutional policy.

US andexanet sales have ceased; conditional EU/UK availability requires benefit/thrombosis/cost review. Traumatic antiplatelet-associated ICH has no uniform platelet-transfusion recommendation; do not use automatically.

Check three/four-factor products, units, registration and availability; use local alternatives if4F is unavailable without equating FFP with4F-PCC. Shared decisions and thrombosis monitoring apply. Pay particular attention to early thrombosis within72 hours after andexanet.

Chinese shock2023

Chinese perioperative

European bleeding2023

FDA2025

ACEP2026

NCS/SCCM2026

BSH2026

Follow institutional policy. Medical education and supervised training.

Pregnancy trauma: maternal/obstetric care

Uterus at/above umbilicus; unsensitized Rh-negative abdominal/major trauma.

Mother first; left uterine displacement; Rh D

At least20 weeks: manual uterine displacement or left tilt15–30 degrees; unsensitized RhD-negative at least13+0 weeks: KB-guided anti-D assessment and availability check; test HCG when appropriate (Chinese R20). Follow institutional policy.

Instability, pain/bleeding or fetal concern: trauma/obstetric escalation.

No equivalent Chinese guidance for these specific items was verified; international reference, local obstetric confirmation.

Queensland pregnancy

Chinese shock2023

Follow institutional policy. Medical education and supervised training.

Pregnancy arrest: resuscitative delivery

Fundus at/above umbilicus (approximately at least20 weeks), no ROSC with standard resuscitation; skilled team.

Prepare at arrest recognition; goal delivery by ≤5 min Follow institutional policy.

AHA 2025: organize obstetric/anesthetic/neonatal teams and resuscitative delivery at the arrest site. Follow institutional policy.

Four-minute initiation/five-minute birth remains the AHA2025 timing goal; call/prepare at arrest recognition, perform on site; benefit may remain beyond five minutes, so do not abandon solely for delay; address traumatic causes. Follow institutional policy.

Four/five-minute timing does not justify waiting four minutes to call, prepare equipment or neonatal rescue; maternal resuscitation and delivery preparation are parallel.

AHA2025

Queensland pregnancy

Follow institutional policy. Medical education and supervised training.

Children: weight and first aliquot

RCH major pediatric trauma with poor circulation; assess bleeding in parallel.

Measured/recent weight; crystalloid 10 mL/kg Follow institutional policy.

Suspected bleeding with compromise: small initial resuscitation aliquots and early blood; RCH initial TXA15 mg/kg, with maximum/rate checked in the pediatric product protocol. Follow institutional policy.

Reassess each aliquot; ongoing hemorrhage needs blood-bank/pediatric MTP, avoiding repeated large crystalloids.

Chinese major-hemorrhage component regimen is on the next card; routine RBC table excludes emergency major bleeding.

RCH: major pediatric trauma primary survey

WS/T795—2022

Follow institutional policy. Medical education and supervised training.

Children: Chinese hemorrhagic transfusion index

WS/T795—2022 §9.5.2.2: pediatric traumatic hemorrhagic shock.

RBC : FFP = 1 : 1 (volume) Follow institutional policy.

Standard example: RBC 20 mL/kg + FFP 20 mL/kg; not routine anemia/non-trauma ratio. Follow institutional policy.

With cumulative RBC above40 mL/kg, consider platelets15–20 mL/kg and cryo10 mL/kg; adjust to Hb above80 g/L, Fib above1.5 g/L, APTT/PT below1.5 times reference midpoint and PLT above75×10⁹/L; monitor overload. Follow institutional pediatric policy. Follow institutional policy.

A component-volume regimen, at a different stage from RCH initial aliquots; check product/local pediatric policy.

WS/T795—2022

Pediatric interpretation2025

Follow institutional policy. Medical education and supervised training.

Burns: Chinese rule of nines

Head/face/neck; both arms; trunk including perineum; both legs including buttocks/feet.

Adult: 9 + 18 + 27 + 46 = 100% Follow institutional policy.

Under 12: head/neck=9+(12−age)%; both legs=46−(12−age)%. Follow institutional policy.

IV resuscitation assessment: adult partial/full-thickness over15%, child over5%, elder over10% or full-thickness over5%; inhalation injury/physiology may independently escalate. Follow institutional policy.

2025 standard Table 1 / §2.1; fluid-formula area uses second/third degree, excluding superficial burns.

NHC burns2025

Follow institutional policy. Medical education and supervised training.

Burns: Chinese adult fluid formula

Chinese 2025 crystalloid/colloid formula; second/third degree; enter percentage number.

First 24 h: 1.5–2 mL × kg × %TBSA + 2000 mL Follow institutional policy.

Crystalloid:colloid=1–2:1; half estimated total in first eight hours from injury, quarters in next two intervals. Follow institutional policy.

Urine0.5–1 mL/(kg·h),1–2 for electrical/crush/hemolytic chemical injury; second-day crystalloid/colloid half first-day actual amount, maintenance unchanged; allocate maintenance per burn service. Follow institutional policy.

1.5 is the lower end, not a fixed universal amount; include prior fluids and follow local burn policy.

NHC burns2025

Follow institutional policy. Medical education and supervised training.

Resuscitation, consent and handover records

Only when urgent rescue prevented timely writing; not permission to delay all records.

Truthful, marked supplementation ≤6 h after rescue ends Follow institutional policy.

Minute-level time, actions, responses, staff, communication and handover ownership.

Record signing difficulties and use lawful emergency authorization; never fabricate signatures/times.

Record norms Article 22(8), 25/27; transfusion consent also follows 2025 blood-use standards.

Basic norms for medical record writing (2010)

NHC clinical blood use technical standard (2025)

Follow institutional policy. Medical education and supervised training.

Trauma bay: before arrival

Tiered recall; resident key-step simulation requires supervision and local policy.

Roles: head / chest / circulation / recorder

Confirm space, monitors, blood bank and senior support

Handover: mechanism, findings, actions, unknowns

Escalate immediate threats or inadequate resources

WHO trauma care checklist and team handover

Chinese prehospital2026

Follow institutional policy. Medical education and supervised training.

xABCDE: revisit each cycle

Tiered recall; resident key-step simulation requires supervision and local policy.

x bleeding → A airway/protection → B breathing → C circulation → D neurology → E exposure/warmth

Specific findings + trends + owner

Check effect after actions; revisit unknown regions

Deterioration: return to primary survey and escalate

Chinese prehospital2026

NICE NG39

Follow institutional policy. Medical education and supervised training.

Pelvis: parallel risk recognition

Tiered recall; resident key-step simulation requires supervision and local policy.

WSES: SBP below90 or maintained by ongoing fluids/blood/vasopressors; BD over6, SI over1 or4–6 RBC units in24 hours also indicate risk. Do not directly convert national component units. Follow institutional policy.

Center the binder at greater trochanters/symphysis level; knees together and internal leg rotation when associated injuries allow. Avoid repeated springing. Coordinate blood, warming, TXA and other-source checks. Follow institutional policy.

Binder→arterial bleeding without laparotomy need: AE; laparotomy/rapid control need: preperitoneal packing plus external fixation. ZoneIII REBOA is a skilled-team bridge in extreme instability with suspected pelvic source, not a mandatory final step. Follow institutional policy.

WSES pelvic2017

NICE NG37

European bleeding2023

Chinese shock2023

Care chain: continuity at transitions

Tiered recall; resident key-step simulation requires supervision and local policy.

Confirm destination capability and owner

Hand over knowns / unknowns / concerns / next review

Before movement check monitoring, airway, bleeding, equipment

Bottleneck or patient change: reassess destination

WHO trauma care checklist and team handover

Chinese prehospital2026

Follow institutional policy. Medical education and supervised training.

Disaster: patient and system review

Tiered recall; resident key-step simulation requires supervision and local policy.

Walking initially green; apnea after airway opening black; RR over30, absent radial pulse/CRT over two seconds, or inability to obey commands red; otherwise yellow. Reassess initial tags. Follow institutional policy.

ERBP2012: early pre-extrication saline, adult1000 mL/h/child15–20 mL/kg/h; after two hours adult at most500/child about10 mL/kg/h or lower. Do not mechanically apply to bleeding/TBI/cardiac-renal failure. Follow institutional policy.

Primary pressure wave(lung/ear/hollow organs), secondary fragments, tertiary displacement, quaternary burns/inhalation/crush. Lung signs may appear up to48 hours; isolated eardrum rupture neither measures nor excludes serious injury. Follow institutional policy.

START adult/HHS

JumpSTART pediatric/HHS

SALT/MUCC/HHS

Chinese medical-rescue plan2006

ERBP crush2012

Follow institutional policy. Medical education and supervised training.

Resuscitation: parallel tasks

Tiered recall; resident key-step simulation requires supervision and local policy.

Hemorrhage control and resource coordination

Early TXA / blood / coagulation monitoring

Warmth, iCa, perfusion and response checks

See separate adult/TBI numerical cards

Chinese shock2023

European bleeding2023

Follow institutional policy. Medical education and supervised training.

Secondary / tertiary: prevent missed injury

Tiered recall; resident key-step simulation requires supervision and local policy.

After stabilization: AMPLE + head-to-toe exam

Mark unassessed regions, pending imaging and limits

Repeat exam, reconcile final reports, assign ownership

New symptoms/deterioration: xABCDE review without waiting

RCH secondary survey

WHO trauma care checklist and team handover

Follow institutional policy. Medical education and supervised training.

Special populations: identify group

Tiered recall; resident key-step simulation requires supervision and local policy.

Older adults: anticoagulation, serious low-energy injury, baseline

Children: weight, age, blood protocol

Pregnancy: mother first, obstetric collaboration

Burns: depth, Chinese rule of nines, specialist review

Chinese prehospital2026

WS/T795—2022

NHC burns2025

Queensland pregnancy

Follow institutional policy. Medical education and supervised training.

Handover: record and accept

Tiered recall; resident key-step simulation requires supervision and local policy.

Findings/time; actions/response

Unknowns and review deadlines

Communication / consent / receiver confirmation

See record card for six-hour rescue supplementation Follow institutional policy.

Basic norms for medical record writing (2010)

NHC clinical blood use technical standard (2025)

Follow institutional policy. Medical education and supervised training.

Pelvic MDT case: staged debrief

Tiered recall; resident key-step simulation requires supervision and local policy.

Early IV antibiotics; NICE open-fracture goal within one hour. Protect perineal wounds/document contamination; trained digital rectal assessment when rectal injury suspected, proctoscopy if positive. Follow institutional policy.

Debridement/packing and blood resuscitation proceed together. Fecal diversion or suprapubic drainage depends on rectal/urinary injury, not routine for everyone. Proposed teaching pathways are not documented performed care. Follow institutional policy.

Do not create a plausible real trajectory when original vitals/Hb/lactate/BE are missing. Any simulation must be distinctly synthetic. Follow institutional policy.

WSES pelvic2017

NICE NG37

Chinese shock2023

Skills: decision to reassessment

Tiered recall; resident key-step simulation requires supervision and local policy.

Timing → preparation/authorization → key steps

Effect checks → failure recognition → escalation

0 absent / 1 after prompting / 2 complete independently Follow institutional policy.

Supervised simulation; scores do not authorize practice

NICE NG39

Follow institutional policy. Medical education and supervised training.

TBI: neurology with ABC

Tiered recall; resident key-step simulation requires supervision and local policy.

Severe TBI: SBP at least100 at50–69 years, at least110 at15–49 or over70; Chinese MAP at least80 mmHg. CPP60–70, treat ICP over22; BTF pressure evidence LevelIII, ICP/CPP LevelIIB. Follow institutional policy.

Chinese SpO_2 floor90%, ideally over95%; ACS target at least94%. Without intracranial hypertension, $PaCO_2$/$PetCO_2$ reference35–45 mmHg; shock invalidates assuming $EtCO_2$ equals arterial CO_2. Follow institutional policy.

Chinese recommendation22: within three hours,1 g slow IV over ten minutes, then1 g over at least eight hours. NICE: age at least16, GCS at most12, no active extracranial bleeding: consider a single2 g IV dose within two hours, before imaging. Follow institutional policy.

Chinese adult TBI2025

BTF severe TBI4

NICE NG232

Follow institutional policy. Medical education and supervised training.

Chest: bilateral comparison

Tiered recall; resident key-step simulation requires supervision and local policy.

Adults: fourth/fifth interspace just anterior to midaxillary line; second interspace midclavicular alternative. Children: second midclavicular. At least8 cm,14G/10G is an adult TCCC equipment reference; adjust to anatomy/device. Follow institutional policy.

Instability or active thoracic bleeding over1500 mL or over200 mL/h for three hours needs immediate surgical evaluation. Stable at least500 mL: drain; below300 mL: close observation;300–499 mL requires individualized assessment. Follow institutional policy.

Pneumothorax8.5–24 Fr; EAST conditionally supports14 Fr or smaller pigtails in stable hemothorax, while WSES-AAST favors28–40 Fr for larger hemothorax. Unstable patients need large-bore drainage; populations differ. Follow institutional policy.

Chinese prehospital2026

WSES-AAST thoracic2025

EAST hemothorax2021

Abdomen: consider occult bleeding

Tiered recall; resident key-step simulation requires supervision and local policy.

SBP below90 with shock signs, or pressure requiring ongoing fluids/blood/vasopressors, indicates instability. Positive FAST with no response goes directly to surgery/damage-control hemorrhage control; responders may have monitored contrast CT. Follow institutional policy.

Interpret BE sign with lactate and trend; do not use the WSES printed BE greater than−5 as a stand-alone trigger or interchange BD with negative BE. Follow institutional policy.

WSES I: stable AAST I–II;II: stable AAST III;III: stable AAST IV–V;IV: unstable at any anatomic grade. Follow institutional policy.

NICE NG39

WSES liver2020

WSES spleen2017

Polytrauma: global priorities

Tiered recall; resident key-step simulation requires supervision and local policy.

Synthetic EFAST addition: absent left pleural sliding, right sliding present, no clear pericardial fluid. With reduced left sounds and BP78/48, trained teams decompress/drain and reassess without CT delay. Follow institutional policy.

Arrival SI126/86 about1.47, ABC at least2: activate assessment for the institutional massive-transfusion pathway while controlling the source. Follow institutional policy.

Major bleeding Fib at most1.5 g/L prompts replacement assessment; European iCa1.1–1.3 mmol/L, Chinese at least1.00. Given Fib1.7/iCa1.05 prompt review, not false claims that every replacement threshold is crossed. Follow institutional policy.

Chinese shock2023

European bleeding2023

WSES-AAST thoracic2025

Follow institutional policy. Medical education and supervised training.

Abdominal case: debrief trends

Tiered recall; resident key-step simulation requires supervision and local policy.

Original SBP67 with hypoxemia warrants escalation before58. Relative bradycardia prompts beta-blocker/anticoagulant history but does not negate shock. Original FAST, age, Hb, lactate, BE and GCS components remain unknown. Follow institutional policy.

Conditional exercise: unstable positive FAST with no response→damage-control laparotomy; suspected pelvis→binder/packing pathway. Prioritize blood and check TXA timing; do not describe proposed care as performed. Follow institutional policy.

NICE NG39

European bleeding2023

WSES pelvic2017

Shock reassessment: improvement?

Tiered recall; resident key-step simulation requires supervision and local policy.

Original FAST records negative abdomen/pleura, not a negative pericardium. Add subxiphoid view, pump function and right-heart assessment early in persistent shock, not at round8. Hb and contrast-CT status are unconfirmed. Follow institutional policy.

For suspected hemorrhage prioritize blood, limit crystalloid; SBP80–90 is a reference before control without severe TBI. Do not rewrite original pressurized fluid as confirmed transfusion. Follow institutional policy.

Anaphylaxis with circulatory/respiratory compromise: adult adrenaline0.5 mg IM, repeat after five minutes if needed per local pathway. AMPLE A asks allergies; improvement after steroids neither proves causality nor substitutes first-line adrenaline. Follow institutional policy.

Chinese shock2023

European bleeding2023

RCUK special-circumstances resuscitation (2025)

Spine/hip: neurovascular recheck

Tiered recall; resident key-step simulation requires supervision and local policy.

Bilateral arm burning pain/weakness with relative leg sparing raises central-cord concern. Complete ASIA/ISNCSCI motor/sensory/sacral assessment, not symptom-only diagnosis. Ejection/paresthesia are high risk: SMR and CT, then MRI for cord abnormalities. Follow institutional policy.

Reduce urgently; six hours is a research comparison reference, not permission to wait. Document sciatic/peroneal function and perfusion before/after; post-reduction CT assesses fracture/fragments/congruence, followed by AVN surveillance. Follow institutional policy.

Sedation needs dedicated monitoring, waveform CO_2, resuscitation/airway rescue and maintained SMR with anesthesia/orthopedic roles. ACEP2026 recommends correcting volume depletion first where feasible. Follow institutional policy.

NICE NG41

NICE NG37

Hip reduction timing2017

ACEP sedation2026

Follow institutional policy. Medical education and supervised training.

Orthopedics: systemic then local

Tiered recall; resident key-step simulation requires supervision and local policy.

Early IV antibiotics, ideally within one hour. Chinese2019 GustiloI–II first-generation cephalosporin(cefazolin), III adds aminoglycoside, fecal/anaerobic contamination adds penicillin/metronidazole; allergy/renal function/local resistance guide alternatives. Follow institutional policy.

High risk: hourly exam, pain/passive stretch/analgesic dose. Symptoms: release circumferential layers to skin, heart-level limb, reassess within30 minutes and consult orthopedics immediately. Follow institutional policy.

Absent pulse/ongoing bleeding/expanding hematoma are hard signs; CRT/Doppler alone cannot exclude injury. Immediate orthopedic/vascular/plastic coordination, temporary shunt when needed. Follow institutional policy.

Chinese open fracture2019

BOAST open fracture2017

NICE NG37

BOAST compartment2025

BOAST arterial2026

Spine: protection and serial neurology

Tiered recall; resident key-step simulation requires supervision and local policy.

Active MAP augmentation lower limit75–80/upper90–95 mmHg for3–7 days, weak recommendation/very-low evidence. Do not lower every spontaneous pressure above95; distinguish severe-TBI targets. Follow institutional policy.

When feasible decompress within24 hours; combine neurologic findings, CT-then-MRI, complete ASIA/sacral assessment. Follow institutional policy.

AANS/CNS2013 discourages routine MPSS; AO2017 weakly offers adult24-hour infusion starting within eight hours as an option, not after eight or48-hour infusion. Specialists/local policy weigh benefit/infection risk. Follow institutional policy.

AO/Praxis SCI2024

AO SCI steroid2017

NICE NG41

Follow institutional policy. Medical education and supervised training.

Upper limb: pre/post comparison

Tiered recall; resident key-step simulation requires supervision and local policy.

Early IV antibiotics, ideally within one hour. Chinese2019 GustiloI–II first-generation cephalosporin(cefazolin), III adds aminoglycoside, fecal/anaerobic contamination adds penicillin/metronidazole; allergy/renal function/local resistance guide alternatives. Follow institutional policy.

High risk: hourly exam, pain/passive stretch/analgesic dose. Symptoms: release circumferential layers to skin, heart-level limb, reassess within30 minutes and consult orthopedics immediately. Follow institutional policy.

Absent pulse/ongoing bleeding/expanding hematoma are hard signs; CRT/Doppler alone cannot exclude injury. Immediate orthopedic/vascular/plastic coordination, temporary shunt when needed. Follow institutional policy.

Chinese open fracture2019

BOAST open fracture2017

NICE NG37

BOAST compartment2025

BOAST arterial2026

Lower limb: local and systemic risk

Tiered recall; resident key-step simulation requires supervision and local policy.

Early IV antibiotics, ideally within one hour. Chinese2019 GustiloI–II first-generation cephalosporin(cefazolin), III adds aminoglycoside, fecal/anaerobic contamination adds penicillin/metronidazole; allergy/renal function/local resistance guide alternatives. Follow institutional policy.

High risk: hourly exam, pain/passive stretch/analgesic dose. Symptoms: release circumferential layers to skin, heart-level limb, reassess within30 minutes and consult orthopedics immediately. Follow institutional policy.

Absent pulse/ongoing bleeding/expanding hematoma are hard signs; CRT/Doppler alone cannot exclude injury. Immediate orthopedic/vascular/plastic coordination, temporary shunt when needed. Follow institutional policy.

Chinese open fracture2019

BOAST open fracture2017

NICE NG37

BOAST compartment2025

BOAST arterial2026

Bvm: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Assess breathing/protection/SMR, suction blood/secretions, choose appropriate oral/nasal adjunct without contraindications. Follow institutional policy.

Two-person two-hand seal/jaw support; assistant bags, checking chest rise and leak. Follow institutional policy.

If oxygen/chest rise fails check seal, obstruction, position and chest causes; persistent failure escalates airway with SpO_2/$EtCO_2$/perfusion review. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

BTS emergency oxygen guideline (2017)

RCUK special-circumstances resuscitation (2025)

NICE NG41

AHA adult BLS2025

Intubation: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

DAS adult video laryngoscopy first; use trauma-specific SMR/MILS, suction, B/C/D backup plans and assigned roles. Follow institutional policy.

JTS severe-hemorrhage anesthesia references reduced/titrated ketamine about1 mg/kg IV, propofol0.5–1 mg/kg IV with hypotension risk, not default doses for everyone; local anesthesia assessment. Follow institutional policy.

Confirm with sustained waveform CO_2, secure and check depth/bilateral lungs/SpO_2/pressure; reconfirm after movement, enter DAS rescue on failure. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

DAS intubation2025

JTS trauma anesthesia2021/2023

NICE NG41

Cricothyrotomy: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

DAS adult #10 scalpel, bougie,6.0 mm cuffed tube, full paralysis/suction. Maximal extension belongs to its adult difficult-airway context; suspected trauma spine injury requires oxygen-rescue/SMR judgment, not routine forced extension. Follow institutional policy.

Trainer: caudal-to-cephalad vertical midline incision up to about8 cm, finger identify membrane; transverse stab, scalpel perpendicular, rotate90 degrees edge caudal. Follow institutional policy.

Inflate cuff/connect ventilation, sustained waveform CO_2; failure considers smaller tube/extended incision/position or operator change, then secure/handover. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

DAS intubation2025

NICE NG41

Needle Decompression: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Adult fourth/fifth interspace just anterior to midaxillary; alternate second midclavicular, not medial to nipple; child second midclavicular. Follow institutional policy.

Adult TCCC14G/10G, about8 cm(3.25 inches) catheter reference; local anatomy/device selection, perpendicular over superior border of the lower rib. Follow institutional policy.

Persistent failure or skilled hospital setting: open/finger thoracostomy then drain; TCCC post-second-attempt field care differs from the hospital pathway. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

TCCC2024

Chinese prehospital2026

NICE NG39

WSES-AAST thoracic2025

Tube Thoracostomy: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Fourth–sixth interspace anterior-to-midaxillary safe triangle, confirm patient anatomy/imaging; trained blunt dissection/finger confirmation, no blind trocar. Follow institutional policy.

Pneumothorax8.5–24 Fr; stable hemothorax EAST conditional14 Fr pigtail, larger/unstable WSES28–40 Fr(no added benefit above32). Follow institutional policy.

Output over1500 or over200 mL/h for three hours, or ongoing instability prompts immediate surgical escalation; physiology may trigger before volume, review retained blood/blockage. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

WSES-AAST thoracic2025

EAST hemothorax2021

NICE NG39

Efast: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Bilateral pleura: sliding/lung point/fluid; negative/absent sliding have alternatives, not a single-image diagnosis. Follow institutional policy.

Report windows/quality/findings/missing views/time. Follow institutional policy.

Repeat matched to physiology; monitored stable responders get contrast CT, suspected tension is treated without completing ultrasound first. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

NICE NG39

RCH: major pediatric trauma primary survey

WSES pelvic2017

Tourniquet: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Visible wound: TCCC5–7 cm proximal, off a joint; Chinese reference at least10 cm per selected system. If unseen, high-and-tight; tighten/add a second alongside/proximal if ineffective. Follow institutional policy.

Confirm both bleeding cessation and absent distal pulse, with pulse not applicable to amputation; pulse alone is insufficient. Follow institutional policy.

Aim for conversion within two hours only without shock, with close monitoring, and not amputation; after six hours do not remove without monitoring/labs. Trained team conversion only. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

TCCC2024

Chinese prehospital2026

Chinese shock2023

Pelvic Binder: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Center at greater trochanters/symphysis, not iliac crest/waist; check clothing/skin. Follow institutional policy.

Knees together/internal rotation when associated injuries allow; gentle observation, no repeated springing. Follow institutional policy.

Persistent instability→AE/PPP/stabilization; plan removal/replacement, NICE within24 hours with surgical plan for instability. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

WSES pelvic2017

NICE NG37

Chinese shock2023

Follow institutional policy. Medical education and supervised training.

Splinting: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Open fracture: early IV antibiotics within one hour, saline-moistened/occlusive dressing, authorized photo, no ED irrigation/probing; local tetanus policy. Follow institutional policy.

Femoral-shaft traction depends on device/pattern; avoid with pelvis or ipsilateral knee/lower-leg/ankle injury, use other immobilization. Follow institutional policy.

Suspected compartment syndrome: release circumferential dressings/splint to skin, limb at heart level, immediate orthopedics; a pulse does not exclude it. Follow institutional policy.

OSCE0 omitted/unsafe,1 prompted,2 independent; total10, critical failure overrides. Follow institutional policy.

NICE NG37

NICE NG41

Hypothermia and damage control

Adult major traumatic bleeding; use a core-temperature method

Core below35 C; rewarm toward36–37 C Follow institutional policy.

Remove wet clothing, insulate and warm patient/infusions/blood;37 C laboratory testing may understate hypothermic coagulopathy Follow institutional policy.

T at most35 C with pH at most7.2, coagulopathy, ongoing bleeding and lengthy-reconstruction risk: consider staged damage-control surgery. Follow institutional policy.

Chinese R12/R29; one temperature does not decide surgery alone

Chinese shock2023

European bleeding2023

Follow institutional policy. Medical education and supervised training.

Pediatric burn fluids

Children; maintenance70–100 mL/kg, under two years100–150 mL/kg

First24 hours:2 mL×kg×%TBSA (partial/full thickness)+maintenance Follow institutional policy.

Crystalloid:colloid1–3:1; half in first eight hours from injury, remainder in the next two eight-hour periods; urine about1 mL/(kg·h). Follow institutional burn policy. Follow institutional policy.

Over5% may cause shock; head/face/neck injury raises airway concern; adjust to previous fluids, circulation and glucose. Follow institutional policy.

Do not use adult2000 mL maintenance; use actual weight and percentage-number area

NHC burns2025

Follow institutional policy. Medical education and supervised training.